

2					CCCCCCCCC	00000000000	BBBBBBBBBBB	00000000000	LL		2	
4					CC	00	00	BB	BB	00	00	LL
6					CC	00	00	BB	BB	00	00	LL
8					CC	00	00	BB	BB	00	00	LL
10					CC	00	00	BB	BB	00	00	LL
12					CC	00	00	BB	BB	00	00	LL
14					CC	00	00	BB	BB	00	00	LL
16					CC	00	00	BB	BB	00	00	LL
18					CC	00	00	BB	BB	00	00	LL
20					CC	00	00	BB	BB	00	00	LL
22					CC	00	00	BB	BB	00	00	LL
24					CC	00	00	BB	BB	00	00	LL
26					CC	00	00	BB	BB	00	00	LL
28					CC	00	00	BB	BB	00	00	LL
30					CC	00	00	BB	BB	00	00	LL
32					CC	00	00	BB	BB	00	00	LL
34					CC	00	00	BB	BB	00	00	LL
36					CC	00	00	BB	BB	00	00	LL
38					CC	00	00	BB	BB	00	00	LL
40					CC	00	00	BB	BB	00	00	LL
42					CC	00	00	BB	BB	00	00	LL
44					CC	00	00	BB	BB	00	00	LL
46					CC	00	00	BB	BB	00	00	LL
48					CC	00	00	BB	BB	00	00	LL
50					CC	00	00	BB	BB	00	00	LL
52					CC	00	00	BB	BB	00	00	LL
54					CC	00	00	BB	BB	00	00	LL
56					CC	00	00	BB	BB	00	00	LL
58					CC	00	00	BB	BB	00	00	LL
60					CC	00	00	BB	BB	00	00	LL
62					CC	00	00	BB	BB	00	00	LL
64					CC	00	00	BB	BB	00	00	LL
16					JJJJJJJJJ	11	444	444	444	AAAAA		
18					JJJJJJJJJ	111	4444	4444	4444	AAAAA		
20					JJ	1111	44 44	44 44	44 44	AA	AA	
22					JJ	11	44 44	44 44	44 44	AA	AA	
24					JJ	11	44 44	44 44	44 44	AA	AA	
26					JJ	11	444444444444	444444444444	444444444444	AAAAA		
28					JJ	11	444444444444	444444444444	444444444444	AAAAA		
30					JJ	11	44	44	44	AA	AA	
32					JJ	11	44	44	44	AA	AA	
34					JJ	11	44	44	44	AA	AA	
36					JJ	11	44	44	44	AA	AA	
38					JJ	11	44	44	44	AA	AA	
40					JJ	11	44	44	44	AA	AA	
42					JJ	11	44	44	44	AA	AA	
44					JJ	11	44	44	44	AA	AA	
46					JJ	11	44	44	44	AA	AA	
48					JJ	11	44	44	44	AA	AA	
50					JJ	11	44	44	44	AA	AA	
52					JJ	11	44	44	44	AA	AA	
54					JJ	11	44	44	44	AA	AA	
56					JJ	11	44	44	44	AA	AA	
58					JJ	11	44	44	44	AA	AA	
60					JJ	11	44	44	44	AA	AA	
62					JJ	11	44	44	44	AA	AA	
64					JJ	11	44	44	44	AA	AA	
30	****A	START	JOB 1444	COBOL	COBOL PROGRAM	ROOM	2.36.02 PM 14 JUN 26	PRINTER1	SYS HMVS	JOB 1444	START	A****

J E S 2 J O B L O G

14.35.44 JOB 1444 \$HASP373 COBOL STARTED - INIT 1 - CLASS A - SYS HMVS
14.35.44 JOB 1444 IEF403I COBOL - STARTED - TIME=14.35.44
14.35.44 JOB 1444 IEC130I SYSPUNCH DD STATEMENT MISSING
14.35.44 JOB 1444 IEC130I SYSLIB DD STATEMENT MISSING
14.35.44 JOB 1444 IEC130I SYSPUNCH DD STATEMENT MISSING
14.35.44 JOB 1444 IEFACTRT COB /IKFCBL00/00:00:00.09/00:00:00.24/00000/COBOL
14.35.45 JOB 1444 IEFACTRT GO /LOADER /00:00:00.03/00:00:00.09/00000/COBOL
14.35.45 JOB 1444 IEF404I COBOL - ENDED - TIME=14.35.45
14.35.45 JOB 1444 \$HASP395 COBOL ENDED

----- JES2 JOB STATISTICS -----

14 JUN 26 JOB EXECUTION DATE

174 CARDS READ

324 SYSOUT PRINT RECORDS

0 SYSOUT PUNCH RECORDS

0.00 MINUTES EXECUTION TIME

```

CCCCCCCCCCCC 000000000000 BBBB BBBB BBBB 000000000000 LL
CCCCCCCCCCCC 000000000000 BBBB BBBB BBBB 000000000000 LL
CC          CC 00          00 BB          BB 00          00 LL
CC          00          00 BB          BB 00          00 LL
CC          00          00 BB          BB 00          00 LL
CC          00          00 BBBB BBBB          00          00 LL
CC          00          00 BBBB BBBB          00          00 LL
CC          00          00 BB          BB 00          00 LL
CC          00          00 BB          BB 00          00 LL
CC          00          00 BB          BB 00          00 LL
CCCCCCCCCCCC 000000000000 BBBB BBBB BBBB 000000000000 LLLLLLLLLLLLLL
CCCCCCCCCCCC 000000000000 BBBB BBBB BBBB 000000000000 LLLLLLLLLLLLLL

```

JJJJJJJJJ	11	444	444	444	AAAAAAAAA
JJJJJJJJJ	111	4444	4444	4444	AAAAAAAAAAAA
JJ	1111	44 44	44 44	44 44	AA AA
JJ	11	44 44	44 44	44 44	AA AA
JJ	11	44 44	44 44	44 44	AA AA
JJ	11	444444444444	444444444444	444444444444	AAAAAAAAAAAA
JJ	11	4444444444444	4444444444444	4444444444444	AAAAAAAAAAAA
JJ	11	44	44	44	AA AA
JJ JJ	11	44	44	44	AA AA
JJ JJ	11	44	44	44	AA AA
JJJJJJJ	1111111111	44	44	44	AA AA
JJJJJJ	1111111111	44	44	44	AA AA

```

*****A  END  JOB 1444  COBOL      COBOL PROGRAM      ROOM      2.36.02 PM 14 JUN 26  PRINTER1  SYS HMVS  JOB 1444  END  A*****

```

```

CCCCCCCCCCCC 000000000000 BBBBBBBBBBBB 000000000000 LL
CCCCCCCCCCCC 000000000000 BBBBBBBBBBBB 000000000000 LL
CC          CC 00          00 BB          BB 00          00 LL
CC          00          00 BB          BB 00          00 LL
CC          00          00 BB          BB 00          00 LL
CC          00          00 BBBBBBBBBBBB 00          00 LL
CC          00          00 BBBBBBBBBBBB 00          00 LL
CC          00          00 BB          BB 00          00 LL
CC          00          00 BB          BB 00          00 LL
CC          00          00 BB          BB 00          00 LL
CCCCCCCCCCCC 000000000000 BBBBBBBBBBBB 000000000000 LLLLLLLLLLLLLL
CCCCCCCCCCCC 000000000000 BBBBBBBBBBBB 000000000000 LLLLLLLLLLLLLL

```

JJJJJJJJJ	11	444	444	444	AAAAAAAA
JJJJJJJJJ	111	4444	4444	4444	AAAAAAAAAAAA
JJ	1111	44 44	44 44	44 44	AA AA
JJ	11	44 44	44 44	44 44	AA AA
JJ	11	44 44	44 44	44 44	AA AA
JJ	11	444444444444	444444444444	444444444444	AAAAAAAAAAAA
JJ	11	4444444444444	4444444444444	4444444444444	AAAAAAAAAAAA
JJ	11	44	44	44	AA AA
JJ JJ	11	44	44	44	AA AA
JJ JJ	11	44	44	44	AA AA
JJJJJJJ	1111111111	44	44	44	AA AA
JJJJJJ	1111111111	44	44	44	AA AA

```

*****A  START  JOB 1444  COBOL          COBOL PROGRAM          ROOM          2.36.02 PM 14 JUN 26  PRINTER1  SYS HMVS  JOB 1444  START  A*****

```

1	//COBOL JOB 1, 'COBOL PROGRAM', CLASS=A, MSGCLASS=X	JOB 1444	
2	//COBOL EXEC PROC=COBUG		
3	XXCOBUG PROC CPARM1='LOAD',	100000100	2
4	XX CPARM2='SIZE=2048K, BUF=1024K'	00000200	4
4	XXCOB EXEC PGM=IKFCBL00, REGION=4096K,	00000301	
6	XX PARM='&CPARM1, &CPARM2'	00000401	6
5	XXSTEPLIB DD DSN=SYSC.LINKLIB, DISP=SHR	00000501	
6	XXSYSPRINT DD SYSOUT=*	00000600	8
7	XXSYSUT1 DD UNIT=SYSDA, SPACE=(460, (700, 100))	00000700	
8	XXSYSUT2 DD UNIT=SYSDA, SPACE=(460, (700, 100))	00000800	10
9	XXSYSUT3 DD UNIT=SYSDA, SPACE=(460, (700, 100))	00000900	
10	XXSYSUT4 DD UNIT=SYSDA, SPACE=(460, (700, 100))	00001000	12
11	XXSYSLIN DD DSN=&LOADSET, DISP=(MOD, PASS),	00001100	
14	XX UNIT=SYSDA, SPACE=(80, (500, 100))	00001200	14
12	//COB.SYSIN DD *		
13	XXGO EXEC PGM=LOADER, PARM='MAP, LET', COND=(5, LT, COB), REGION=100K	00001300	16
14	XXSYSLIN DD DSN=*.COB.SYSLIN, DISP=(OLD, DELETE)	00001400	
15	XXSYSLOUT DD SYSOUT=*	00001504	18
16	XXSYSLIB DD DSN=SYSC.COBLIB, DISP=SHR	00001602	
17	//GO.INPUT DD *		20
18	//GO.OUTPUT DD SYSOUT=*,		
22	// DCB=(RECFM=FBA, LRECL=133, BLKSIZE=13300)		22
	//		
24			24
26			26
28			28
30			30
32			32
34			34
36			36
38			38
40			40
42			42
44			44
46			46
48			48
50			50
52			52
54			54
56			56
58			58
60			60
62			62
64			64

```

CCCCCCCCCCCC 000000000000 BBBBBBBBBBBB 000000000000 LL
CCCCCCCCCCCC 000000000000 BBBBBBBBBBBB 000000000000 LL
CC          CC 00          00 BB          BB 00          00 LL
CC          00          00 BB          BB 00          00 LL
CC          00          00 BB          BB 00          00 LL
CC          00          00 BBBBBBBBBB 00          00 LL
CC          00          00 BBBBBBBBBB 00          00 LL
CC          00          00 BB          BB 00          00 LL
CC          00          00 BB          BB 00          00 LL
CC          00          00 BB          BB 00          00 LL
CCCCCCCCCCCC 000000000000 BBBBBBBBBBBB 000000000000 LLLLLLLLLLLL
CCCCCCCCCCCC 000000000000 BBBBBBBBBBBB 000000000000 LLLLLLLLLLLL

```

JJJJJJJJJ	11	444	444	444	AAAAAAAA
JJJJJJJJJ	111	4444	4444	4444	AAAAAAAAAAAA
JJ	1111	44 44	44 44	44 44	AA AA
JJ	11	44 44	44 44	44 44	AA AA
JJ	11	44 44	44 44	44 44	AA AA
JJ	11	444444444444	444444444444	444444444444	AAAAAAAAAAAA
JJ	11	4444444444444	4444444444444	4444444444444	AAAAAAAAAAAA
JJ	11	44	44	44	AA AA
JJ JJ	11	44	44	44	AA AA
JJ JJ	11	44	44	44	AA AA
JJJJJJJ	1111111111	44	44	44	AA AA
JJJJJJ	1111111111	44	44	44	AA AA

```

*****A  END  JOB 1444  COBOL      COBOL PROGRAM      ROOM      2.36.02 PM 14 JUN 26  PRINTER1  SYS HMVS  JOB 1444  END  A*****

```

JJJ	11	444	444	444	AAAAAAAA
JJJ	111	4444	4444	4444	AAAAAAAAAAAA
	1111	44 44	44 44	44 44	AA AA
	11	44 44	44 44	44 44	AA AA
	11	44 44	44 44	44 44	AA AA
	11	4444444444	4444444444	4444444444	AAAAAAAAAAAA
	11	444444444444	444444444444	444444444444	AAAAAAAAAAAA
	11	44	44	44	AA AA
	11	44	44	44	AA AA
	11	44	44	44	AA AA
	1111111111	44	44	44	AA AA
	1111111111	44	44	44	AA AA

```

****A  START  JOB 1444  COBOL      COBOL PROGRAM      ROOM      2.36.02 PM 14 JUN 26  PRINTER1  SYS HMVS  JOB 1444  START  A****

```

STMT NO. MESSAGE

```
2 4 IEF653I SUBSTITUTION JCL - PARM='LOAD,SIZE=2048K,BUF=1024K'
4 IEF236I ALLOC. FOR COBOL COB COBOL
6 IEF237I 253 ALLOCATED TO STEPLIB
8 IEF237I 253 ALLOCATED TO SYS00258
10 IEF237I JES2 ALLOCATED TO SYSPRINT
12 IEF237I 191 ALLOCATED TO SYSUT1
14 IEF237I 191 ALLOCATED TO SYSUT2
16 IEF237I 191 ALLOCATED TO SYSUT3
18 IEF237I 191 ALLOCATED TO SYSUT4
20 IEF237I 191 ALLOCATED TO SYSLIN
22 IEF237I JES2 ALLOCATED TO SYSIN
24 IEC130I SYSPUNCH DD STATEMENT MISSING
26 IEC130I SYSLIB DD STATEMENT MISSING
28 IEC130I SYSPUNCH DD STATEMENT MISSING
30 IEF142I COBOL COB COBOL - STEP WAS EXECUTED - COND CODE 0000
32 IEF285I SYSC.LINKLIB KEPT *------0
34 IEF285I VOL SER NOS= SYSCPK.
36 IEF285I UCSYSCPK KEPT *------0
38 IEF285I VOL SER NOS= SYSCPK.
40 IEF285I JES2.JOB01444.S00103 SYSOUT
42 IEF285I SYS26165.T143544.RA000.COBOL.R0000001 DELETED *------6
44 IEF285I VOL SER NOS= MYSTUF.
46 IEF285I SYS26165.T143544.RA000.COBOL.R0000002 DELETED *------6
48 IEF285I VOL SER NOS= MYSTUF.
50 IEF285I SYS26165.T143544.RA000.COBOL.R0000003 DELETED *------9
52 IEF285I VOL SER NOS= MYSTUF.
54 IEF285I SYS26165.T143544.RA000.COBOL.R0000004 DELETED *------3
56 IEF285I VOL SER NOS= MYSTUF.
58 IEF285I SYS26165.T143544.RA000.COBOL.LOADSET PASSED *------83
60 IEF285I VOL SER NOS= MYSTUF.
62 IEF285I JES2.JOB01444.SI0101 SYSIN
64 IEF373I STEP /COB / START 26165.1435
66 IEF374I STEP /COB / STOP 26165.1435 CPU 0MIN 00.07SEC SRB 0MIN 00.02SEC VIRT 2076K SYS 228K
68 **** JOB NAME: COBOL JOBCARD READ 2026/165 14:35:44 370/ 33 VS2 R03.8 HMVS *****
70 *
72 * STEP NUMBER: 1 USER CORE: 2076K START TIME: 14:35:44 CPU TIME: 00:00:00.09 ACTIVE TIME: 00:00:00.17 *
74 * STEP NAME: COB SYSTEM CORE: 228K STOP TIME: 14:35:44 SRB TIME: 00:00:00.02 ALLOC TIME: 14:35:44 *
76 * PROGRAM NAME: IKFCBL00 REGION SIZE: 4096K ELAPSED TIME: 00:00:00.24 TCB TIME: 00:00:00.07 PROGRAM LOAD: 14:35:44 *
78 * CONDITION CODE: 00000 PERFORMANCE GROUP: 004 *
80 * JES2 CARDS: 14 SERVICE UNITS PAGES IN/OUT # SWAPS PAGES SWAP IN/OUT VIO PAGES IN/OUT *
82 * 828 0 / 0 0 0 / 0 0 / 0 *
84 *
86 * ADDR/UNIT I/O COUNT ADDR/UNIT I/O COUNT ADDR/UNIT I/O COUNT ADDR/UNIT I/O COUNT ADDR/UNIT I/O COUNT ADDR/UNIT I/O COUNT *
88 * 253/D3350 0 253/D3350 0 191/D3390 6 191/D3390 6 191/D3390 9 191/D3390 3 *
90 * 191/D3390 83 *
92 *****
94 IEF236I ALLOC. FOR COBOL GO COBOL
96 IEF237I 191 ALLOCATED TO SYSLIN
98 IEF237I JES2 ALLOCATED TO SYSLOUT
100 IEF237I 253 ALLOCATED TO SYSLIB
102 IEF237I 253 ALLOCATED TO SYS00260
104 IEF237I JES2 ALLOCATED TO INPUT
106 IEF237I JES2 ALLOCATED TO OUTPUT
108 IEF142I COBOL GO COBOL - STEP WAS EXECUTED - COND CODE 0000
110 IEF285I SYS26165.T143544.RA000.COBOL.LOADSET DELETED *------84
112 IEF285I VOL SER NOS= MYSTUF.
114 IEF285I JES2.JOB01444.S00104 SYSOUT
116 IEF285I SYSC.COBLIB KEPT *------12
118 IEF285I VOL SER NOS= SYSCPK.
```



```
IEF285I UCSYSCPK                                KEPT          *-----0
2 IEF285I VOL SER NOS= SYSCPK.
IEF285I JES2.JOB01444.SI0102
4 IEF285I JES2.JOB01444.S00105                      SYSIN
IEF373I STEP /GO          / START 26165.1435          SYSOUT
6 IEF374I STEP /GO          / STOP 26165.1435 CPU      0MIN 00.02SEC SRB      0MIN 00.01SEC VIRT  100K SYS  220K
*****
8 *
* STEP NUMBER:          2  USER CORE:          100K  START TIME:   14:35:44    CPU TIME:    00:00:00.03  ACTIVE TIME:  00:00:00.05 *
10 * STEP NAME:          GO    SYSTEM CORE:       220K  STOP TIME:   14:35:45    SRB TIME:    00:00:00.01  ALLOC TIME:   14:35:44 *
* PROGRAM NAME:  LOADER    REGION SIZE:       100K  ELAPSED TIME: 00:00:00.09  TCB TIME:    00:00:00.02  PROGRAM LOAD: 14:35:44 *
12 * CONDITION CODE:   00000  PERFORMANCE GROUP: 004
* JES2 CARDS:          2          SERVICE UNITS  PAGES IN/OUT  # SWAPS  PAGES SWAP IN/OUT  VIO PAGES IN/OUT *
14 *                               585          0 /    0          0          0 /    0          0 /    0
*
* ADDR/UNIT I/O COUNT  ADDR/UNIT I/O COUNT  ADDR/UNIT I/O COUNT  ADDR/UNIT I/O COUNT  ADDR/UNIT I/O COUNT  ADDR/UNIT I/O COUNT
16 * 191/D3390          84 253/D3350          12 253/D3350          0
*****
18 IEF375I JOB /COBOL    / START 26165.1435
20 IEF376I JOB /COBOL    / STOP 26165.1435 CPU      0MIN 00.09SEC SRB      0MIN 00.03SEC
22
24
26
28
30
32
34
36
38
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42
44
46
48
50
52
54
56
58
60
62
64
```

JJJ	11	444	444	444	AAAAAAAA
JJJ	111	4444	4444	4444	AAAAAAAAAAAA
	1111	44 44	44 44	44 44	AA AA
	11	44 44	44 44	44 44	AA AA
	11	44 44	44 44	44 44	AA AA
	11	4444444444	4444444444	4444444444	AAAAAAAAAAAA
	11	444444444444	444444444444	444444444444	AAAAAAAAAAAA
	11	44	44	44	AA AA
	11	44	44	44	AA AA
	11	44	44	44	AA AA
	1111111111	44	44	44	AA AA
	1111111111	44	44	44	AA AA

```
*****A  END  JOB 1444  COBOL      COBOL PROGRAM      ROOM      2.36.02 PM 14 JUN 26  PRINTER1  SYS HMVS  JOB 1444  END  A*****
```

```

CCCCCCCCCCCC 000000000000 BBBBBBBBBBBB 000000000000 LL
CCCCCCCCCCCC 000000000000 BBBBBBBBBBBB 000000000000 LL
CC          CC 00          00 BB          BB 00          00 LL
CC          00          00 BB          BB 00          00 LL
CC          00          00 BB          BB 00          00 LL
CC          00          00 BBBBBBBBBBBB 00          00 LL
CC          00          00 BBBBBBBBBBBB 00          00 LL
CC          00          00 BB          BB 00          00 LL
CC          00          00 BB          BB 00          00 LL
CC          00          00 BB          BB 00          00 LL
CCCCCCCCCCCC 000000000000 BBBBBBBBBBBB 000000000000 LLLLLLLLLLLLLL
CCCCCCCCCCCC 000000000000 BBBBBBBBBBBB 000000000000 LLLLLLLLLLLLLL

```

JJJJJJJJJ	11	444	444	444	AAAAAAAAA
JJJJJJJJJ	111	4444	4444	4444	AAAAAAAAAAAA
JJ	1111	44 44	44 44	44 44	AA AA
JJ	11	44 44	44 44	44 44	AA AA
JJ	11	44 44	44 44	44 44	AA AA
JJ	11	444444444444	444444444444	444444444444	AAAAAAAAAAAA
JJ	11	4444444444444	4444444444444	4444444444444	AAAAAAAAAAAA
JJ	11	44	44	44	AA AA
JJ JJ	11	44	44	44	AA AA
JJ JJ	11	44	44	44	AA AA
JJJJJJJ	1111111111	44	44	44	AA AA
JJJJJJ	1111111111	44	44	44	AA AA

```
*****A  START  JOB 1444  COBOL          COBOL PROGRAM          ROOM          2.36.02 PM 14 JUN 26  PRINTER1  SYS HMVS  JOB 1444  START  A*****
```

1

```
00001 IDENTIFICATION DIVISION.
00002 PROGRAM-ID. GRADE-LIST2.
00003 *
00004 ENVIRONMENT DIVISION.
00005 CONFIGURATION SECTION.
00006 SOURCE-COMPUTER. IBM-370.
00007 OBJECT-COMPUTER. IBM-370.
00008 *
00009 INPUT-OUTPUT SECTION.
00010 FILE-CONTROL.
00011     SELECT GRADE-FILE ASSIGN TO UT-S-INPUT.
00012     SELECT PRINT-FILE ASSIGN TO UT-S-OUTPUT.
00013 *
00014 DATA DIVISION.
00015 FILE SECTION.
00016 FD  GRADE-FILE
00017     LABEL RECORD IS OMITTED.
00018 01  GRADE-RECORD.
00019     05 STUDENT-ID-IN          PIC X(10).
00020     05 STUDENT-NAME-IN       PIC X(25).
00021     05 FILLER                PIC X(05).
00022     05 STUDENT-SCORE1-IN     PIC 99.
00023     05 FILLER                PIC X(03).
00024     05 STUDENT-SCORE2-IN     PIC 99.
00025     05 FILLER                PIC X(03).
00026     05 STUDENT-GRADE-IN      PIC X.
00027     05 FILLER                PIC X(29).
00028 *
00029 FD  PRINT-FILE
00030     LABEL RECORD IS OMITTED.
00031 01  PRINT-LINE                PIC X(132).
00032 *
00033 WORKING-STORAGE SECTION.
00034 77 TOTAL-CARDS                PIC 99      VALUE ZERO.
00035 77 TOTAL-SCORE1               PIC 9(4)    VALUE ZERO.
00036 77 TOTAL-SCORE2               PIC 9(4)    VALUE ZERO.
00037 77 IS-EOF                    PIC X       VALUE 'N'.
00038 *
00039 01 HEADER-LINE.
00040     03 FILLER                  PIC X(43)   VALUE SPACES.
00041     03 FILLER                  PIC X(21)   VALUE
00042         'STUDENT GRADE SUMMARY'.
00043 *
00044 01 TITLE-LINE.
00045     03 FILLER                  PIC X(20)   VALUE SPACES.
00046     03 FILLER                  PIC X(12)   VALUE
00047         'STUDENT NAME'.
00048     03 FILLER                  PIC X(18)   VALUE SPACES.
00049     03 FILLER                  PIC X(09)   VALUE
00050         'ID NUMBER'.
00051     03 FILLER                  PIC X(12)   VALUE SPACES.
00052     03 FILLER                  PIC X(06)   VALUE
00053         'SCORES'.
00054     03 FILLER                  PIC X(04)   VALUE SPACES.
```

```

00055          03 FILLER          PIC X(05)  VALUE
00056          'TOTAL'.
*
00057          01 SUBTITLE-LINE.
00058          03 FILLER          PIC X(71)  VALUE SPACES.
00059          03 FILLER          PIC X       VALUE '1'.
00060          03 FILLER          PIC X(05)  VALUE SPACES.
00061          03 FILLER          PIC X       VALUE '2'.
00062          03 FILLER          PIC X(03)  VALUE SPACES.
00063          03 FILLER          PIC X(05)  VALUE 'SCORE'.
00064          03 FILLER          PIC X(07)  VALUE ' GRADE'.
00065          *
00066          01 DETAIL-LINE.
00067          03 FILLER          PIC X(20)  VALUE SPACES.
00068          03 DETAIL-NAME      PIC X(25) .
00069          03 FILLER          PIC X(05)  VALUE SPACES.
00070          03 DETAIL-ID       PIC X(10) .
00071          03 FILLER          PIC X(10)  VALUE SPACES.
00072          03 DETAIL-SCORE-1  PIC 99 .
00073          03 FILLER          PIC X(04)  VALUE SPACES.
00074          03 DETAIL-SCORE-2  PIC 99 .
00075          03 FILLER          PIC X(04)  VALUE SPACES.
00076          03 DETAIL-TOTAL-SCORE PIC Z99 .
00077          03 FILLER          PIC X(05)  VALUE SPACES.
00078          03 DETAIL-GRADE    PIC X .
00079          *
00080          01 SUMMARY-LINE.
00081          03 FILLER          PIC X(35)  VALUE SPACES.
00082          03 FILLER          PIC X(20)  VALUE
00083          'AVERAGE SCRORES FOR '.
00084          03 SUMMARY-CARDS-READ PIC 99    VALUE ZERO.
00085          03 FILLER          PIC X(13)  VALUE
00086          ' STUDENTS: '.
00087          03 SUMMARY-SCORE-1  PIC 99    VALUE ZERO.
00088          03 FILLER          PIC X(04)  VALUE SPACES.
00089          03 SUMMARY-SCORE-2  PIC 99    VALUE ZERO.
00090          03 FILLER          PIC X(04)  VALUE SPACES.
00091          03 SUMMARY-TOTAL-SCORE PIC Z99  VALUE ZERO.
00092          *
00093          PROCEDURE DIVISION.
00094          000-MAIN.
00095          PERFORM INITIALIZATION THRU INITIALIZATION-EXIT.
00096          PERFORM READ-AND-PRINT THRU READ-AND-PRINT-EXIT
00097          UNTIL IS-EOF = 'Y' OR TOTAL-CARDS = 99.
00098          PERFORM PRINT-SUMMARY.
00099          PERFORM CLOSING.
00100          STOP RUN.
*
00101          INITIALIZATION.
00102          OPEN INPUT GRADE-FILE.
00103          OPEN OUTPUT PRINT-FILE.
00104          WRITE PRINT-LINE FROM HEADER-LINE BEFORE 2 LINES.
00105          WRITE PRINT-LINE FROM TITLE-LINE BEFORE 2 LINES.
00106          WRITE PRINT-LINE FROM SUBTITLE-LINE AFTER 1 LINES.
00107          READ GRADE-FILE
00108          AT END MOVE 'Y' TO IS-EOF.
00109          INITIALIZATION-EXIT.

```

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2
4 00112      EXIT.
00113      *
6 00114      READ-AND-PRINT.
00115          MOVE STUDENT-NAME-IN TO DETAIL-NAME.
8 00116          MOVE STUDENT-ID-IN TO DETAIL-ID.
00117          MOVE STUDENT-SCORE1-IN TO DETAIL-SCORE-1.
10 00118          MOVE STUDENT-SCORE2-IN TO DETAIL-SCORE-2.
00119          ADD STUDENT-SCORE1-IN, STUDENT-SCORE2-IN
12 00120              GIVING DETAIL-TOTAL-SCORE.
00121          MOVE STUDENT-GRADE-IN TO DETAIL-GRADE.
14 00122          ADD 1 TO TOTAL-CARDS.
00123          ADD STUDENT-SCORE1-IN TO TOTAL-SCORE1.
16 00124          ADD STUDENT-SCORE2-IN TO TOTAL-SCORE2.
00125          WRITE PRINT-LINE FROM DETAIL-LINE AFTER 1 LINES.
18 00126          READ GRADE-FILE
00127              AT END MOVE 'Y' TO IS-EOF.
20 00128      READ-AND-PRINT-EXIT.
00129      EXIT.
22 00130      *
00131      PRINT-SUMMARY.
24 00132          MOVE TOTAL-CARDS TO SUMMARY-CARDS-READ.
00133          IF TOTAL-CARDS > 0 THEN
26 00134              DIVIDE TOTAL-CARDS INTO TOTAL-SCORE1
00135                  GIVING SUMMARY-SCORE-1 ROUNDED
28 00136              DIVIDE TOTAL-CARDS INTO TOTAL-SCORE2
00137                  GIVING SUMMARY-SCORE-2 ROUNDED.
30 00138      *      END-IF.
00139          ADD SUMMARY-SCORE-1, SUMMARY-SCORE-2
32 00140              GIVING SUMMARY-TOTAL-SCORE.
00141          WRITE PRINT-LINE FROM SUMMARY-LINE AFTER 2 LINES.
34 00142      *
00143      CLOSING.
36 00144          CLOSE GRADE-FILE.
00145          CLOSE PRINT-FILE.
```

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```
*STATISTICS*      SOURCE RECORDS = 145      DATA DIVISION STATEMENTS = 60      PROCEDURE DIVISION STATEMENTS = 34
*OPTIONS IN EFFECT*  SIZE = 2097152  BUF = 1048576  LINECNT = 57  SPACE1, FLAGW, SEQ, SOURCE
*OPTIONS IN EFFECT*  NODMAP, NOPMAP, NOCLIST, SUPMAP, NOXREF, LOAD, NODECK, APOST, NOTRUNC, NOLIB, NOVERB
*OPTIONS IN EFFECT*  ZWB
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2				CCCCCCCCC	00000000000	BBBBBBBBBBB	00000000000	LL		2		
4				CC	00	00	BB	00	00	4		
6				CC	00	00	BB	00	00	6		
8				CC	00	00	BB	00	00	8		
10				CC	00	00	BB	00	00	10		
12				CC	00	00	BB	00	00	12		
14				CC	00	00	BB	00	00	14		
16				CC	00	00	BB	00	00	16		
18				CC	00	00	BB	00	00	18		
20				CC	00	00	BB	00	00	20		
22				CC	00	00	BB	00	00	22		
24				CC	00	00	BB	00	00	24		
26				CC	00	00	BB	00	00	26		
28				CC	00	00	BB	00	00	28		
30				CC	00	00	BB	00	00	30		
32				CC	00	00	BB	00	00	32		
34				CC	00	00	BB	00	00	34		
36				CC	00	00	BB	00	00	36		
38				CC	00	00	BB	00	00	38		
40				CC	00	00	BB	00	00	40		
42				CC	00	00	BB	00	00	42		
44				CC	00	00	BB	00	00	44		
46				CC	00	00	BB	00	00	46		
48				CC	00	00	BB	00	00	48		
50				CC	00	00	BB	00	00	50		
52				CC	00	00	BB	00	00	52		
54				CC	00	00	BB	00	00	54		
56				CC	00	00	BB	00	00	56		
58				CC	00	00	BB	00	00	58		
60				CC	00	00	BB	00	00	60		
62				CC	00	00	BB	00	00	62		
64				CC	00	00	BB	00	00	64		
16		JJJJJJJJJ	11	444	444	444	AAAAA			16		
18		JJJJJJJJJ	111	4444	4444	4444	AAAAA			18		
20		JJ	1111	44 44	44 44	44 44	AA	AA		20		
22		JJ	11	44 44	44 44	44 44	AA	AA		22		
24		JJ	11	44 44	44 44	44 44	AA	AA		24		
26		JJ	11	444444444444	444444444444	444444444444	AAAAA			26		
28		JJ	11	444444444444	444444444444	444444444444	AAAAA			28		
30		JJ	11	44	44	44	AA	AA		30		
32		JJ	11	44	44	44	AA	AA		32		
34		JJ	11	44	44	44	AA	AA		34		
36		JJ	11	44	44	44	AA	AA		36		
38		JJ	11	44	44	44	AA	AA		38		
40		JJ	11	44	44	44	AA	AA		40		
42		JJ	11	44	44	44	AA	AA		42		
44		JJ	11	44	44	44	AA	AA		44		
46		JJ	11	44	44	44	AA	AA		46		
48		JJ	11	44	44	44	AA	AA		48		
50		JJ	11	44	44	44	AA	AA		50		
52		JJ	11	44	44	44	AA	AA		52		
54		JJ	11	44	44	44	AA	AA		54		
56		JJ	11	44	44	44	AA	AA		56		
58		JJ	11	44	44	44	AA	AA		58		
60		JJ	11	44	44	44	AA	AA		60		
62		JJ	11	44	44	44	AA	AA		62		
64		JJ	11	44	44	44	AA	AA		64		
30	****A	END	JOB 1444	COBOL	COBOL PROGRAM	ROOM	2.36.02 PM 14 JUN 26	PRINTER1	SYS HMVS	JOB 1444	END	A****


```

CCCCCCCCCCCC 000000000000 BBBBBBBBBBBB 000000000000 LL
CCCCCCCCCCCC 000000000000 BBBBBBBBBBBB 000000000000 LL
CC          CC 00          00 BB          BB 00          00 LL
CC          00          00 BB          BB 00          00 LL
CC          00          00 BB          BB 00          00 LL
CC          00          00 BBBBBBBBBB 00          00 LL
CC          00          00 BBBBBBBBBB 00          00 LL
CC          00          00 BB          BB 00          00 LL
CC          00          00 BB          BB 00          00 LL
CC          00          00 BB          BB 00          00 LL
CCCCCCCCCCCC 000000000000 BBBBBBBBBBBB 000000000000 LLLLLLLLLLLL
CCCCCCCCCCCC 000000000000 BBBBBBBBBBBB 000000000000 LLLLLLLLLLLL

```

JJJJJJJJJ	11	444	444	444	AAAAAAAAA
JJJJJJJJJ	111	4444	4444	4444	AAAAAAAAAAAA
JJ	1111	44 44	44 44	44 44	AA AA
JJ	11	44 44	44 44	44 44	AA AA
JJ	11	44 44	44 44	44 44	AA AA
JJ	11	444444444444	444444444444	444444444444	AAAAAAAAAAAA
JJ	11	4444444444444	4444444444444	4444444444444	AAAAAAAAAAAA
JJ	11	44	44	44	AA AA
JJ JJ	11	44	44	44	AA AA
JJ JJ	11	44	44	44	AA AA
JJJJJJJ	1111111111	44	44	44	AA AA
JJJJJJ	1111111111	44	44	44	AA AA

```
*****A  START  JOB 1444  COBOL          COBOL PROGRAM          ROOM          2.36.02 PM 14 JUN 26  PRINTER1  SYS HMVS  JOB 1444  START  A*****
```

VS LOADER

```
OPTIONS USED - PRINT, MAP, LET, CALL, RES, NOTERM, SIZE=69632, NAME=**GO
```

NAME	TYPE	ADDR	NAME	TYPE	ADDR	NAME	TYPE	ADDR	NAME	TYPE	ADDR	NAME	TYPE	ADDR
------	------	------	------	------	------	------	------	------	------	------	------	------	------	------

8	GRADE0LI	SD	9C010	ILB0STP0*	SD	9CC68	ILB0STP1*	LR	9CC7E	ILB0SPA0*	SD	9CCA0	8
---	----------	----	-------	-----------	----	-------	-----------	----	-------	-----------	----	-------	---

10	TOTAL LENGTH	F68	10
----	--------------	-----	----

ENTRY	ADDRESS	9C010

```

CCCCCCCCCCCC 000000000000 BBBBBBBBBBBB 000000000000 LL
CCCCCCCCCCCC 000000000000 BBBBBBBBBBBB 000000000000 LL
CC          CC 00          00 BB          BB 00          00 LL
CC          00          00 BB          BB 00          00 LL
CC          00          00 BB          BB 00          00 LL
CC          00          00 BBBBBBBBBB 00          00 LL
CC          00          00 BBBBBBBBBB 00          00 LL
CC          00          00 BB          BB 00          00 LL
CC          00          00 BB          BB 00          00 LL
CC          00          00 BB          BB 00          00 LL
CCCCCCCCCCCC 000000000000 BBBBBBBBBBBB 000000000000 LLLLLLLLLLLL
CCCCCCCCCCCC 000000000000 BBBBBBBBBBBB 000000000000 LLLLLLLLLLLL

```

JJJJJJJJJ	11	444	444	444	AAAAAAAA
JJJJJJJJJ	111	4444	4444	4444	AAAAAAAAAAAA
JJ	1111	44 44	44 44	44 44	AA AA
JJ	11	44 44	44 44	44 44	AA AA
JJ	11	44 44	44 44	44 44	AA AA
JJ	11	444444444444	444444444444	444444444444	AAAAAAAAAAAA
JJ	11	4444444444444	4444444444444	4444444444444	AAAAAAAAAAAA
JJ	11	44	44	44	AA AA
JJ JJ	11	44	44	44	AA AA
JJ JJ	11	44	44	44	AA AA
JJJJJJJ	1111111111	44	44	44	AA AA
JJJJJJ	1111111111	44	44	44	AA AA

```

*****A  END  JOB 1444  COBOL      COBOL PROGRAM      ROOM      2.36.02 PM 14 JUN 26  PRINTER1  SYS HMVS  JOB 1444  END  A*****

```

```

CCCCCCCCCCCC 000000000000 BBBBBBBBBBBB 000000000000 LL
CCCCCCCCCCCC 000000000000 BBBBBBBBBBBB 000000000000 LL
CC          CC 00          00 BB          BB 00          00 LL
CC          00          00 BB          BB 00          00 LL
CC          00          00 BB          BB 00          00 LL
CC          00          00 BBBBBBBBBB 00          00 LL
CC          00          00 BBBBBBBBBB 00          00 LL
CC          00          00 BB          BB 00          00 LL
CC          00          00 BB          BB 00          00 LL
CC          00          00 BB          BB 00          00 LL
CCCCCCCCCCCC 000000000000 BBBBBBBBBBBB 000000000000 LLLLLLLLLLLL
CCCCCCCCCCCC 000000000000 BBBBBBBBBBBB 000000000000 LLLLLLLLLLLL

```

JJJJJJJJJ	11	444	444	444	AAAAAAAAA
JJJJJJJJJ	111	4444	4444	4444	AAAAAAAAAAAA
JJ	1111	44 44	44 44	44 44	AA AA
JJ	11	44 44	44 44	44 44	AA AA
JJ	11	44 44	44 44	44 44	AA AA
JJ	11	444444444444	444444444444	444444444444	AAAAAAAAAAAA
JJ	11	4444444444444	4444444444444	4444444444444	AAAAAAAAAAAA
JJ	11	44	44	44	AA AA
JJ JJ	11	44	44	44	AA AA
JJ JJ	11	44	44	44	AA AA
JJJJJJJ	1111111111	44	44	44	AA AA
JJJJJJ	1111111111	44	44	44	AA AA

```
*****A  START  JOB 1444  COBOL          COBOL PROGRAM          ROOM          2.36.02 PM 14 JUN 26  PRINTER1  SYS HMVS  JOB 1444  START  A*****
```

STUDENT GRADE SUMMARY						
2	STUDENT NAME	ID NUMBER	SCORES		TOTAL	
4						
6			1	2	SCORE	GRADE
8	JOSHUA WATSON	447-221	48	68	116	A
8	LAUREN RODRIGUEZ	123-456	27	33	60	B
	NATHAN SCOTT	654-852	63	55	118	C
10	BENJAMIN PEREZ	159-753	73	87	160	D
	ASHLEY WARD	612-782	60	65	125	E
12	EVELYN BELL	963-852	18	14	32	F
	GABRIEL ANDERSON	854-698	25	33	58	A
14	TIMOTHY CARTER	654-852	23	28	51	B
	OLIVER MOORE	112-254	31	44	75	C
16	OWEN WOOD	787-584	55	65	120	D
	ROBERT ROSS	446-187	42	52	94	E
18	HEATHER THOMAS	268-471	38	58	96	F
	MEGAN SANDERS	784-885	55	60	115	A
20	AUBREY COLLINS	456-753	75	88	163	B
	JAMES WASHINGTON	574-698	47	63	110	C
22	SIMON SULSER	456-587	48	48	96	D
	JAMES BROWN	875-965	66	66	132	E
24	KEVIN BLUE	212-236	52	87	139	F
	HUGH GRANT	985-654	95	55	150	A
26	JAMES KING	265-121	25	43	68	B
28	AVERAGE SCORES FOR 20 STUDENTS:		48	56	104	
30						
32						
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```

CCCCCCCCCCCC 000000000000 BBBB BBBB BBBB 000000000000 LL
CCCCCCCCCCCC 000000000000 BBBB BBBB BBBB 000000000000 LL
CC          CC 00          00 BB          BB 00          00 LL
CC          00          00 BB          BB 00          00 LL
CC          00          00 BB          BB 00          00 LL
CC          00          00 BBBB BBBB          00          00 LL
CC          00          00 BBBB BBBB          00          00 LL
CC          00          00 BB          BB 00          00 LL
CC          00          00 BB          BB 00          00 LL
CC          00          00 BB          BB 00          00 LL
CCCCCCCCCCCC 000000000000 BBBB BBBB BBBB 000000000000 LLLLLLLLLLLL
CCCCCCCCCCCC 000000000000 BBBB BBBB BBBB 000000000000 LLLLLLLLLLLL

```

JJJJJJJJJ	11	444	444	444	AAAAAAAA
JJJJJJJJJ	111	4444	4444	4444	AAAAAAAAAAAA
JJ	1111	44 44	44 44	44 44	AA AA
JJ	11	44 44	44 44	44 44	AA AA
JJ	11	44 44	44 44	44 44	AA AA
JJ	11	444444444444	444444444444	444444444444	AAAAAAAAAAAA
JJ	11	444444444444	444444444444	444444444444	AAAAAAAAAAAA
JJ	11	44	44	44	AA AA
JJ JJ	11	44	44	44	AA AA
JJ JJ	11	44	44	44	AA AA
JJJJJJJ	1111111111	44	44	44	AA AA
JJJJJJ	1111111111	44	44	44	AA AA

```

*****A  END  JOB 1444  COBOL      COBOL PROGRAM      ROOM      2.36.02 PM 14 JUN 26  PRINTER1  SYS HMVS  JOB 1444  END  A*****

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